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Simplicity Studio from Silicon Labs, Design Suits from Xilinx and many others include support for the latest processors focusing on IoT developments and designs.

But apart from these applications, an all-new SQL Planner has been introduced to ease data analysis and management for intelligent devices. SQL Planner from ITTIA is included in its database engine which calculates critical data statistics efficiently with cost-based optimisations. It provides a smart query optimiser that determines the most efficient way to execute a given query by considering plausible query plans, thus accelerating SQL query execution and reducing the time to access data.

SQL Planner can be an important embedded data management tool to help devices gain intelligence in machine-to-machine (M2M) frameworks and thus IoT ecosystems. These devices are enabled to analyse data and determine when objects must be replaced, repaired or recalled.

Automatic NEON vectorisation

NEON is a single instruction multiple data (SIMD) architecture extension implemented as part of ARM processor. But it has its own execution pipelines and a distinct register bank. If you are working with applications based on ARM Cortex-A processors, especially with multimedia and signal processing applications, the support for NEON vectorisation becomes crucial.

IAR embedded workbench has

introduced the possibility to automatically vectorise code, enabling you to achieve faster application response time, improved application battery lifetime and the other requirements of low cost and low power.

Educational hardware

Low-cost target hardware platforms like Arduino, Raspberry Pi and others have become popular among designers as well as education community. This popularity has forced the design tools to have support for these interesting designing blocks. This new feature will not only help the designer community but the education sector as well.

Simulink, which offers block diagram environment for model-based design and simulation, now includes support for prototyping, testing and running models on such low-cost target hardware as Arduino, Lego Mindstorms NXT and Raspberry Pi in its latest version. The latest version also supports Samsung Galaxy Android devices.

For real-time applications

Also, Simulink product family now has a new member, Simulink Real-Time. It allows you to create real-time applications from different models and run them on dedicated as well as custom target computer hardware connected to your physical system.

With the existing support for real-time simulation and testing, rapid control prototyping, DSP and vision system prototyping, and hardware-in-the-loop (HIL) simulations, you can extend your

Simulink models with driver blocks, automatically generate real-time applications, define instrumentation, and perform interactive or automated runs on a dedicated target computer equipped with a real-time kernel, multicore CPU, I/O and protocol interfaces, and FPGAs.

EDA tools

With growing demand for compact and efficient systems, it is getting difficult to have everything fitted inside a box and at the same time maintain reasonable cost. The size constraint can be met with the help of rigid-flex PCB technology, though most tools do not include folding and fitting aspects of rigid-flex designs.

To eliminate these constraints and reduce manual work needed for designing rigid-flex PCBs, the latest version of Altium Designer is equipped with support for flex and rigid-flex designs, including schematic capture, 3D PCB layout, analysis and programmable design—all in a single unified platform. This allows smaller packaging of electronic designs, leading to lower cost of materials and production, and increased durability as well.

A single-layer-stack PCB tool cannot support rigid-flex designs, as both rigid and flex sections may use different sets of layers. The new enhanced layer stack management system supports the definition and naming of multiple stacks to support this design requirement. It allows definition and naming of primary and sub-stacks on a circuit board for use in a rigid-flex circuit design.

While Altium has enhanced its layer-stack management system for supporting rigid-flex designs, OrCAD has been updated with IDX (incremental data exchange) enhancements to support flex circuit definitions. Also, you can import flex circuit bend areas from MCAD in the latest version of OrCAD. ●

The author wrote this as a technical journalist at EFY. He recently opted and shifted to the hands-on training division of EFY